Krish Agarwal

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Research Interests

Efficient ML systems: structured and sparse attention, GPU kernels, serving, and agent infrastructure.

Education

Carnegie Mellon University Aug 2025 – May 2030 (expected)
Ph.D. in Electrical and Computer Engineering, advised by Prof. Beidi Chen Pittsburgh, PA
GPA: 4.0/4.0; Carnegie Institute of Technology Dean's Fellow

The University of Texas at Austin Aug 2022 – May 2025
B.S. in Computer Science, Turing Scholars Honors Program Austin, TX
GPA: 4.0/4.0; graduated with Highest Honors; Distinguished College Scholar
Honors thesis: *Tensor Core Accelerated Kernels for Hadamard and Fourier Masked Attention* [PDF] [Code]
Thesis advisor: Prof. Keshav Pingali

Publications

Conference Papers * Equal contribution

[C2] **MonarchRT: Efficient Attention for Real-Time Video Generation**
K. Agarwal, Z. Chen, C. Luo, Y. Chen, H. Zheng, X. Huang, A. Rudra, B. Chen
Advances in Neural Information Processing Systems (NeurIPS), 2026. [arXiv] [Code] [Blog]

[C1] **L3M+P: Lifelong Planning with Large Language Models**
K. Agarwal*, Y. Jiang*, J. Hu, B. Liu, P. Stone
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025. [arXiv] [Code]

Preprints and Manuscripts Under Review

[P4] **Singularity-Bench: Can Agents Build Agents?**
Y. Zhou*, **K. Agarwal***, Z. Chen, Z. Zhang, W. Li, S. Wu, B. Yuan, B. Chen
Under review, 2026.

[P3] **FlashRT: Agent Harness for Guiding Agents to Deploy Real-Time Multimodal Applications**
K. Agarwal, Z. Chen, Y. Qin, Z. Gu, A. Rudra, B. Chen
Under review, 2026. [arXiv] [Code] [Blog]

[P2] **Efficient MoE Serving in the Memory-Bound Regime: Balance Activated Experts, Not Tokens**
Y. Yu*, H. Ma*, **K. Agarwal**, N. Oswald, Q. Huang, H. Linsenmaier, C. Mei, R. Zhao, R. Borkar, B. Darvish Rouhani, D. Nellans, R. Krashinsky, A. Khandelwal
Preprint, 2025. [arXiv]

[P1] **HadaCore: Tensor Core Accelerated Hadamard Transform Kernel**
K. Agarwal*, R. Astra*, A. Hoque, M. Srivatsa, R. Ganti, L. Wright, S. Chen
Preprint, 2024. [arXiv] [Code] [PyTorch Blog]

Research Experience

Carnegie Mellon University Aug 2025 – Present
Graduate Research Assistant

- Developed MonarchRT, a Monarch matrix attention parameterization for real-time video diffusion that reaches up to 95% attention sparsity with no quality loss on Self-Forcing while enabling real-time generation on a single RTX 5090 [C2].

- Built FlashRT, an agent harness that guides coding agents to turn simple reference implementations of real-time multimodal applications (e.g., video world models, multimodal LLMs) into optimized multi-GPU deployments through a measurement-gated optimization loop, achieving up to $\sim$70$\times$ lower latency and 2.8$\times$/3.6$\times$ higher throughput on NVIDIA B200/AMD MI355X GPUs [P3].
- Co-led Singularity-Bench, an open-ended benchmark measuring how far agents can automate AI development, in which agents build pre-training, post-training, inference, cluster-orchestration, and agent-harness systems that are evaluated against mature human-engineered baselines on identical hardware [P4].

NVIDIA May 2025 – Aug 2025
Research Intern

- Designed a training-free method that predicts Mixture-of-Experts (MoE) routing ahead of time from inter-layer and token information, with up to 90% accuracy on existing MoE models.
- Developed a training-free strategy that cuts activated experts by up to 50% during parallel-sampling test-time scaling, improving compute utilization and weight reuse while maintaining downstream performance.

IBM Research May 2024 – Aug 2024
Software Engineer Intern

- Researched and implemented rotation-based integer quantization for accurate, accelerated LLM inference, improving processing speed by 60–100%.
- Co-developed HadaCore, a Tensor Core-accelerated Hadamard transform kernel that outperforms the prior state of the art by 10–35% depending on problem size; published on the PyTorch blog [P1].

Learning Agents Research Group, UT Austin Jun 2023 – Feb 2024
Undergraduate Research Assistant

- Built a retrieval-augmented LLM framework on an Apache AGE knowledge graph that tracks environment state from natural-language updates and generates PDDL planning goals from retrieved context [C1].

Living with Robots Lab, UT Austin Jan 2023 – Aug 2023
Undergraduate Research Assistant

- Created 3D scene reconstructions supporting semantic queries with Segment Anything (SAM) and CLIP, and improved CLIP accuracy on mislabeled instances through finetuning.

Teaching

Carnegie Mellon University Spring 2026
Teaching Assistant, 18-789: Deep Generative Modeling (graduate course)

Technical Skills

Languages: Python, Java, C/C++, CUDA, Triton, PTX/SASS
Frameworks & Tools: PyTorch, SGLang, vLLM, NVIDIA Nsight Systems/Compute